# Why

The edge of a wildfire is the most dangerous place for a firefighter.

**Robot dogs** (four-legged robots) can climb steep, rocky, smoky ground where trucks get stuck and drones can't see.

> **But a robot near a fire must know where the fire *will* be, not just where it is now.**

# How we predict the fire

**Real fire history**
~246,000 NASA satellite fire detections from 30 major wildfires worldwide

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**Weather**
wind, humidity, temperature, rain

**Terrain**
elevation and slope

**Land cover**
forest, shrub, grass, urban

**Fire shape**
nearby burning cells, fire size

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**Machine learning: XGBoost**
27 inputs per 50 m cell. Trained on 24 fires, tested on 6 fires it never saw.

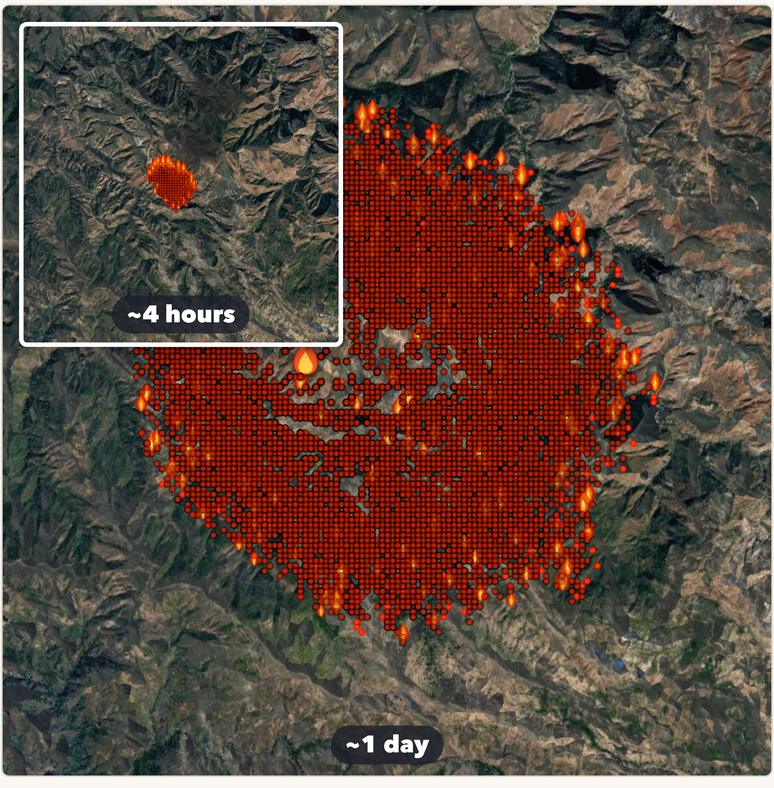
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**Physics: Rothermel spread equation**
fuel × wind × slope × dryness. The ML model adjusts the rate.

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**Output**
the chance each 50 m cell catches fire, every time step

# The simulator




Simulated fire spreading over real 3D terrain in the Big Sur area of Monterey County, California (summer weather). Runs in a web browser on Google 3D map tiles.